Description

Course syllabus for MA 487: Design of Experiments
Winter 2024-25, Department of Mathematics
Rose-Hulman Institute of Technology

**** I reserve the right to modify the course content, schedule, topics, policies, etc. outlined in this syllabus.****

Basic Information

Catalog Description

MA487 is an applied course in design of experiments. Emphasis is placed on designing statistical studies to solve problems in engineering and science. A variety of designs are presented, including the full factorial, screening, response surface, and split plot. It is demonstrated how constraints on the randomization process due to the design are related to the appropriate analysis method and meaning of the results. Statistical software is used for data analysis throughout.

Course Topics

One-, Two-, and M-way ANOVA; Factorial Designs; Multiple Comparisons; Fractional Factorial Designs; Response Surface Methodology; Complete Block Designs; Nested Factor Designs; Incomplete Block Designs; Hasse Diagrams

Prerequisite

MA223 (Engineering Statistics) OR MA382 (Introduction to Statistics)

Role of the Prerequisite

An introductory course in statistics is a requirement for MA 487. We will assume you have experience with statistical inference and terminology. Specifically, we will rely heavily on previous knowledge of hypothesis testing and ANOVA.

No Textbook Required

The class notes and assignments provided within Moodle are comprehensive for the course. Thus, purchase of a textbook is not required. However, if you prefer to have a formal textbook source as a reference, we loosely follow the textbook by Oehlert.

Oehlert. *A First Course in Design and Analysis of Experiments.* (2010). 1st edition. Online pdf available for free.

Statistical Software

We will exclusively use R for data analysis in this course. Please see the 0: Getting Started in MA487 section.

Course Format

Face-to-Face Class Meetings

Monday, Tuesday, Thursday, and Friday @ 11am in G313

It is extremely important that you plan to attend our in-person class sessions. Many of the sessions will be composed of group activity work that is not easy to contribute to virtually.

The Daily Schedule for our class provides an overview of what we will do during each meeting. Each meeting is of a specific type, outlined below.

> **Statistical Inference:** These classes will have more of a "traditional lecture" style. This means that Dr. Heyman will be presenting new content related to statistical methodology relevant to the current module. You'll have notes packets that we fill in during class.
>
> **Experimental Design:** These classes will be a mix of "traditional lecture" and group activities. We will learn new content regarding data collection mechanics and practice with the content via example scenarios.
>
> **Experiment Plan:** These classes are purely a group activity. We will work together to discuss details related to planning a data collection for a controlled experiment, given different process constraints.
>
> **Experiment:** These classes are purely a group activity. You will work together with classmates to implement the process for data collection that was previously planned. There will also be time for groups to work on the report associated with the experiment.

Online Content

Approximately one time per module, completed asynchronously

The Daily Schedule highlights online (ONL) content associated with example data analyses and software demonstration. This content is provided asynchronously online so that you can reference how to implement the statistical analysis technique at any point during the term and work with statistical software at your own pace. There are some students enrolled in the course that may have very little familiarity with using R, while there are others that might classify their knowledge of R programming as expert level.

Important: Although we will not cover this aspect synchronously in person, it is expected that students work through the provided material at some point during the specified day. You will need this material to be able to successfully complete course assignments.

How to Get Help

I'm looking forward to working with you during the coming term! My first recommendation for success is to keep on schedule with the course material. I want our classroom to be a comfortable and engaging atmosphere for learning. If you encounter difficulties, please reach out to me as soon as possible. **I enjoy working with students and anticipate spending time with you as you become proficient and then an expert with our course material throughout the term!**
-Dr. Megan Heyman (heyman@rose-hulman.edu)

Office hours in DL110-B: 9:00am - 10:50am on Tuesday & Friday

- My office is inside the Engineering Management suite. If you enter Moench Hall from the quad entrance at the "commons," turn right down the lower-level hallway. The Engineering Management suite has a glass door on the right side of the hallway. You will need to enter the glass door and then proceed down the passage. My office is the first door on the left of the suite hallway.

Availability other times

- Please feel welcome to come by my office to see if I'm available at other times. I leave my door open almost all the time when I am on campus, and I am on campus a lot. **If my door is open, I am happy to meet with you!**
- I try to check email regularly during the regular work week: 8am-8pm Monday through Friday. *My availability outside of typical hours may be limited.* Although I do use Microsoft Teams, I do not check it as often. **If you cannot find me in my office, email is the best way to contact me quickly.**
- I try to keep my Outlook calendar updated. If you cannot come to in-person office hours or have questions on other days of the week, please do not hesitate to reach out to me over email. **We can set up an ad-hoc meeting either in person or on Teams that works with our schedules.**
- I will make every effort to try to meet with you upon request. I do request your patience and flexibility, as there may be times when I cannot meet with you immediately.

Students with Accessibility Needs

Rose-Hulman is committed to working with students who have special needs or disabilities. Visit the Accessibility Services website for more information. Requests for academic accommodations must be documented with and approved by the Accessibility Services office before they can be implemented in this course.

Student Well-Being and Academic Success

Rose-Hulman cares about the well-being of every student. Stress is a normal part of the college experience and is often compounded both by classwork and unexpected challenges outside the classroom. I encourage you to reflect on how you take care of yourself throughout the term, especially before the demands of exams and projects reach their peak. You can reach out to the Learning Center, Student Academic Success or the Student Counseling Center (which offers both online and in-person resources) if you would like assistance. Additional resources to help maintain your work-life balance can be found on the Campus Life web page and within the Student Success Guide. If you, or someone you know, is struggling, you can let us know through the Rose Cares system, and someone will reach out.

Assessment Structure

You will find that this is NOT a typical mathematics course.

A poor statistical analysis may contain completely valid computations. However, an exemplary statistical analysis not only contains correct mathematical computation, but also, a thorough data exploration and interpretation of results in context.

Additionally, this course will place heavy focus on mechanics of the data collection process and how the way data are collected drive the relevant statistical model. All of our models are related to ANOVA, thus the computation aspect within this course is probably the least interesting part of design of experiments. We'll demonstrate how to use statistical software to fit models in each section, but the process to obtain data and then correctly model the response will be much more important.

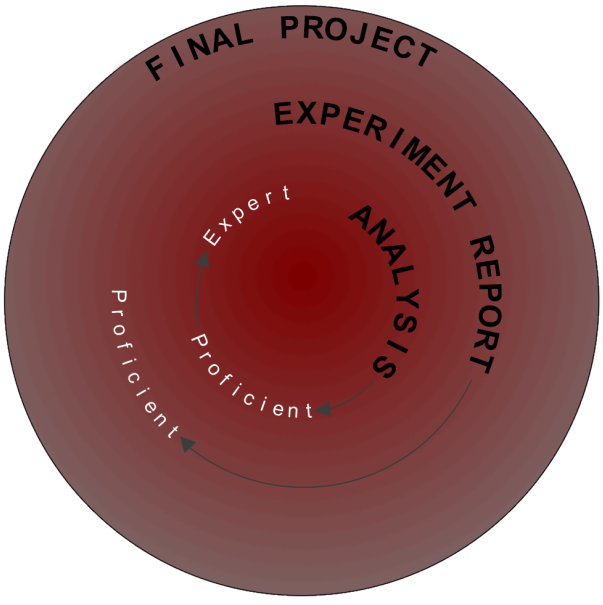
Design of Course Assessments

This course was created using "backwards design." This implies that lectures, activities, and assessments are created based on course goals and objectives.

Assignments are mapped to overall course objectives.

Each class assignment will reference module-level objectives that are assessed. All module-level objectives are mapped to overarching course-level objectives. Thus, the importance or use of any assignment in this class should be clear, since the assignments directly assess valuable objectives.

Structure of Assignment Types



We can visualize our assessments as growth from the core (Analysis assignments) out to the Final Project. We'll focus on growing core skills in design of experiments and once we have the strong basis, incorporate specialized/advanced techniques.

Analysis assignments occur with each module of material. These assignments have an intermediate difficulty level. You will be able to use course resources and feedback to hone your skillset in the module's topic area. Thus, you will have the opportunity to receive guided feedback from Dr. Heyman to revise and resubmit solutions to achieve a "Proficient" or "Expert" rating.

Experiment Reports occur with most modules of material. These assignments are completed as a group, with some questions directly answered during discussion led by Dr. Heyman and others by the group later. These assignments ask students to combine course concepts and several steps of a statistical study within one assessment. Thus, the assignments tend to have a higher difficulty level. Groups will have the opportunity to revise and resubmit solutions to achieve a "Proficient" rating.

The Final Project is optional, depending on the grade you are aiming for. Students aiming for an A, B+, or B will notify Dr. Heyman at the latest, by the end of week 8 of a plan to complete the final project. The final project allows students to gain deeper understanding of an aspect in experimental design that is not covered directly in class. Submissions of the final project are collected during the scheduled finals week slot.

Course Objectives

By the end of this course, students will be able to...

A. **Identify** design components based on a stated scenario.

B. **Recognize** statistical designs based on a stated scenario.

C. **Describe** procedures for implementing specific experimental designs.

D. **Specify** clear directions for a data collection procedure that could be later implemented.

E. **Recommend** an experimental design that incorporates all factors of interest and constraints within the process.

F. **Justify** the choice of an experimental design.

G. **Criticize** an implemented or proposed experimental design.

H. **Use** statistical software to create visualizations of data and reproducible statistical analyses.

I. **Implement** an appropriate statistical analysis based on the implemented study design and available data.

J. **Identify** relevant components of output produced by statistical software.

K. **Discuss** conditions for statistical inference.

L. **Interpret** relevant information from a statistical analysis.

M. **Recommend** improvements for a process (either subsequent experimental design or "best" process settings) using results of a statistical analysis.

N. **Describe** limitations related to the parameter(s) of interest and/or statistical inference technique.

Analysis

There are 8 module analysis assignments.

Each module assignment will consist approximately 5 parts. (Some assignments may have more or fewer parts, but approximately the same time commitment to create solutions is expected for any given assignment.) Each part will be associated with a module level objective and have a set of criteria that you can use as a guide while creating your solution.

All assignments and revisions will be available and collected using Moodle.

Students should aim to submit an attempt of the assignment on time to receive feedback. There is not a penalty to submitting incorrect work since you can use feedback to revise and improve.

Submitting an assignment on time but does not have an attempt at all parts will receive a "not a diligent attempt" rating. Receiving this rating is similar to submitting late work. This means you can revise work to "proficient" level but cannot achieve "expert" level via revision.

A strong grasp of course material and demonstration of engagement is valued.

I want this course to create a strong statistical foundation for you. However, no one is perfect and sometimes the best learning results from first experiencing a difficulty. As such, **students will have the opportunity to revise analysis assignments.**

Any analysis assignment will have the opportunity for up to two revisions during the week after initial submission. This means that if you submit your work on time, you have up to 3 chances to use course resources and feedback to create a solution that is "Proficient" or "Expert" level work.

Dr. Heyman hopes to assignments and revisions on Mondays and Thursdays. Initial submission and revision deadlines are outlined on the course schedule. If Dr. Heyman gets behind with grading, she will notify the class of deadline extensions, as needed.

Students submitting their first attempt of an analysis assignment after the initial deadline (e.g. late submissions and incomplete submissions) can potentially achieve "proficient" level work. **These types of submissions will not be able to revise to achieve "expert" level work.**

Rating of Submitted Solutions

These assignments will be graded on a scale of "not a diligent attempt", "not proficient", "proficient", and "expert".

Expert level assignments meet most, if not all of the provided criteria for all parts. The solution demonstrates that the student has a strong grasp of the material.

Proficient level assignments meet many of the provided criteria. It is clear that the student has a decent grasp of the material, but may be missing some key details.

Not Proficient level assignments indicates that the solution does not demonstrate an understanding of the material or much important work is missing. However, it is clear that the solution attempted to address all parts of the assignment.

Not a Diligent Attempt rated assignments indicates that either an assignment was not submitted, or there was not an attempt of at least one part of the assignment.

Collaboration with classmates is allowed on these assignments. Plagiarism is unacceptable.

It is rare to work completely alone in your job. Students are encouraged to collaborate many times, throughout our course. Analysis assignments are an instance where your learning can be enhanced through collaboration. However, there is a line between collaboration and plagiarism; plagiarism is not beneficial to the knowledge acquisition of any involved party. Please see the course Collaboration Policy about ways you may collaborate without crossing the line.

All students involved in plagiarism will receive a "not proficient" and not be allowed to revise the assignment where plagiarism occurred. In accordance with Mathematics Department policy, Academic Affairs will also be notified of the plagiarism instance. Multiple instances of plagiarism during the term may result in an automatic failing grade.

Use of AI or other resources beyond those provided by Dr. Heyman is addressed in the assignment directions.

Use of resources beyond those provided by Dr. Heyman may be allowed with proper documentation for some of these assignment questions. The directions of the assignment will clearly indicate what resources are allowable and how use of such resources should be documented.

Experiment Report

There are 7 experiment report assignments.

Each module assignment will consist of several questions that your team will answer while planning, implementing, analyzing, and then interpreting an experiment run in class. The questions will be associated with module-level and course-level goals.

All assignments and revisions will be available and collected using Moodle.

Students will work in teams and submit one report per team. Team membership will rotate during the term. Dr. Heyman will assign teams for Report #1, Reports #2&3, Reports #4&5. The class will have the choice of having teams assigned or self-selected for the Reports #6&7. Thus, you will potentially work with four unique groups on these assignments during the term.

Teams should plan to submit complete solutions by the deadline for this assignment. This will provide the most opportunity to revise work, if needed.

A strong grasp of course material and demonstration of engagement is valued.

I want this course to create a strong statistical foundation for you. However, no one is perfect and sometimes the best learning results from first experiencing a difficulty. As such, **teams will have the opportunity to revise experiment reports.**

Any experiment report will have the opportunity for up to two revisions during the week after initial submission. This means that if you submit your work on time, you have up to 3 chances to use course resources and feedback to create a solution that is "Proficient" level work.

Dr. Heyman hopes to assignments and revisions on Mondays and Thursdays. Initial submission and revision deadlines are outlined on the course schedule. If Dr. Heyman gets behind with grading, she will notify the class of deadline extensions, as needed.

Rating of Submitted Solutions

These assignments will be graded on a scale of "not proficient" and "proficient".

Proficient level assignments indicate that the group has a decent grasp of the material, but potentially could be missing be missing some key details.

Not Proficient level assignments indicates that the solution does not demonstrate an understanding of the material or much important work is missing.

Teams submit lab assignments together.

Teams will submit one assignment for the entire group for each report. Thus, everyone in the team should be contributing on any given assignment. Team members will have different levels of proficiency in the course content and different skillsets. As a result, a well-functioning team does not necessarily mean "equal participation" of team members. However, every team member is responsible for the work submitted; therefore, every team member is expected to act professionally - which includes active participation in meetings and aiding in the construction of final documents.

Thus, all participating members of the team are successful on the assignment together. (That is, all participating members on the team receive the same assignment grade.)

Attendance and communication is necessary.

On class days that are experiment days (planning and implementation), attendance is expected and will be tracked. With the expectation that all team members are participating, members of the team that are not participating on the report will not be eligible to receive credit for that assignment.

This does not mean that you should come to class when you are ill and expose everyone else to your illness.

If you are unable to attend on an experiment day, to be eligible to receive credit for the assignment, it is necessary is that you communicate that you are unable to attend. **This is done by emailing your entire team and Dr. Heyman before the absence occurs.** The email keeps accountability within the team and also should indicate how you will still participate in the assignment even though you are physically absent that day. If that participation does not occur, then such students may not be eligible to receive credit on the assignment.

Students emailing that they will miss more than 2 experiment days during the term will need to meet with Dr. Heyman, as the email policy described above does not apply to systematic absence from class.

Use of AI or other resources beyond those provided by Dr. Heyman is addressed in the assignment directions.

Use of resources beyond those provided by Dr. Heyman may be allowed with proper documentation for some of these assignment questions. The directions of the assignment will clearly indicate what resources are allowable and how use of such resources should be documented.

Final Project Report

Students aiming for a grade of "A", "B+", or "B" will complete a Final Project. The project will not be relevant to grades of "C+" or below.

Goal of the Final Project

The goal of this assignment is to ask students aiming for a high achievement letter grade to work beyond the methods directly presented in class on a project within design of experiments that you find interesting. This helps you build out a more advanced skillset, beyond the basics covered in our other course assignments.

You can also use the final project to help build out your "portfolio" of work, which could be especially useful as you think toward graduate school or job interviews. Since this is a specialized project completed beyond the typical undergraduate curriculum, you will have a completed work that you can showcase to potential employers or schools. Some students have even connected this project to supplement research they are conducting outside of our course.

What is the Final Project?

You get to select what type of project you will complete. Although the deadline to notify Dr. Heyman you are completing a project comes at the end of Week 8, it is highly recommended that you start planning for the project much sooner. There are three main types of projects that students can choose from:

Implement Your Own Experiment: Students choosing this type of project will design, implement, and then analyze resulting data from an experiment that they plan. It is possible to work in a team of 2-3 students for this type of project. Students would self-select into the team. Students hoping for this type of project will need to obtain final approval of the plan from Dr. Heyman before implementing the experiment. (We need to ensure that the project proposed follows all relevant safety protocols and human-subject research standards.) This type of project works best for students that have an outside interest that they want to pursue in the design of experiments realm.

Expand/Improve a Course Experiment: Students choosing this type of project will design, implement, and then analyze resulting data from an expanded or improved experiment that we covered during class. It is possible to work in a team of 2-3 students for this type of project. Students would self-select into the team. Students hoping for this type of project will need to work with Dr. Heyman to potentially borrow materials for the data collection. This type of project works best for students that are looking for further design experience and enjoy the applied aspect of this course.

DoE Topic Report: Students choosing this type of project will learn about an advanced topic in design of experiments that we did not formally cover this term. You will write a formal report that explains the advanced topic and either implements a real data analysis using the technique or a set of simulations. Dr. Heyman has a list of several potential topics for students interested in this option. Students work individually on this type of project. This type of project works best for students that are more interested in the theoretical side of design of experiments, perhaps preparing for graduate school.

What to expect with the Final Project

Be sure to notify Dr. Heyman as soon as you have decided that you are planning on completing a final project report. The deadline for the notification is by the end of Week 8, but it is expected that most students have potentially decided on a project before we even leave for winter break. When notifying Dr. Heyman of the intention to complete a project, please specify what type you are planning to complete and potentially if you are working with a team.

After notification, you will communicate with Dr. Heyman regarding progress of the project and questions you have as they arise. The final project report should exhibit a clear, diligent effort to complete the project. **If the project report is clearly not a diligent effort, the course grade might be lowered by half a letter grade.**

> For example, if you have met "A" criteria on Analysis and Experiment Reports but do not diligently complete the project, the assigned course grade will be "B+".

The final project report must be submitted by the end of the slot scheduled during finals week. Late assignments cannot be submitted. There is not a revision opportunity for this assignment, but students that are diligently working and communicating with Dr. Heyman should have no concern that their report will not meet the "diligent effort" criteria.

Grades

I want this course to create a strong foundation in statistical experimental design for you. **In order to help you achieve this, I will be implementing a version of "specifications-based grading."** That is, instead of taking a weighted average of points earned on a series of assignments throughout the term, you will earn grades based on the assignments that you complete at "proficient" and "expert" levels.

Successfully completed assignments ("proficient" or "expert" ratings) earn credit toward your overall course grade. In order to help you achieve a strong rating, very clear expectations will be provided on every assignment, often via a rubric.

You will also notice that I am not expecting perfection. For example, a student may achieve an A in the course via revising work to demonstrate your learning of the topics. You will have the opportunity to revise all course assignments. This awards you for going back to learn the material.

Note that successful solutions will achieve clear, strong standards that are set by listed criteria. The criteria do not translate to percentage points, and definitely don't translate to achieving a 100%. Generally, expert-level ratings are associated with student work near the "B" level, while proficient-level ratings are associated with solutions near a "C" level. **Working consistently throughout the term and implementing feedback to improve will set you up for success in this class.**

Letter Criteria

Final letter grades in this course will be assigned based on your consistent and diligent work to improve your skillset course material throughout the term.

Total possible credits that can be earned.

As a reference, the table below details the total possible number of credits offered via assignments that you can earn during the term.

Assignment Type	Total possible expert credits	Total possible proficient credits
Analysis	8	8
Experiment Report	NA	7

Recall that for any analysis assignment you can achieve either an "expert" or "proficient" level, but not both. The expert level rating asks for stronger solutions than "proficient."

Credits cannot be transferred across assignment types. For example, extra Analysis credits cannot be counted or reassigned into the Experiment Report assignment type.

Letter grade mapping to course credits

The number of credits listed in each column indicates the minimum number of credits associated with the letter.

Assignment Type	Final Grade of "A"	Final Grade of "B"	Final Grade of "C"	Final grade of "D"
Analysis (Expert)	7	5	3	
Analysis (Proficient)	1	2	3	5
Experiment Report	7	6	5	4
Final Project	diligent effort	diligent effort		

To achieve any given letter, you must have at least the minimum number of credits in all of the assignment categories.

> For example, a student that ends the term with 3 expert analyses, 4 proficient analyses, and 7 experiment reports has earned a final letter grade of "C". Notice that the expert-level analysis credits earned meet minimum qualifications for "C" while the other categories have more than the minimum requirements.

Earning "+" Grade

To earn a "+" on the final letter grade, students earn an extra expert analysis credit beyond what is required.

For example, a student that ends the term with 4 expert analyses, 3 proficient analyses, and 7 experiment reports has earned a final letter grade of "C+" since they earned one more expert level credit in the analysis type of assignment than was is required.

Expectations for Our Classroom

Rose-Hulman Institute of Technology is committed to being an inclusive community in which the multiplicity of values, beliefs, intellectual viewpoints, and cultural perspectives enrich learning and inform scholarship. Our classroom will be an open and safe environment to share and discuss ideas. Thus, it is expected that everyone treats each other with mutual respect.

- **Discussion and questions are encouraged during class**. Feel free to speak up at any time during lecture sessions. My classroom is a fairly informal atmosphere.
- If anything occurs during or outside of our class (for example, a microaggression) which makes you feel uncomfortable, please reach out to me. Moreover, if you are a recipient of or a witness to any kind of harassment or bullying, either online or in person, please tell me immediately. **I want all students to feel comfortable in my classroom, and it is my intention to foster equity and inclusion.**
- Lecture attendance is integral to success in this course. It is easy to fall behind from missing 1 or 2 class sessions.
    - Similarly, it is important to complete the online (ONL) material with each module if you plan to have success in this class.

To keep our classroom running efficiently and to remove distractions from our learning goals,

- Please do not use your cell phone during class. It is disruptive to other students and the professor. Laptop use should be responsible and appropriate to our class activities.
- Images need to be professionally appropriate. As noted in the Student Handbook, the display of sexually suggestive objects or pictures constitutes harassment of others who can see the images. Others will see our laptop background pictures, avatars, etc., so we need to choose appropriate images.
- Electronic messages, forum postings, questions, etc. need to be truthful, respectful, and polite. Rose-Hulman's policy for responsible use of computing facilities notes that sending fraudulent, harassing, or obscene messages and/or materials electronically is unethical and may be illegal. Please review the core rules of Netiquette ('Network etiquette,' or rules for polite interactions online) in preparation for this course.
- Humor doesn't always translate well into electronic communications. Something that was intended as funny or sarcastic can be interpreted literally, causing misunderstandings. "Humor" that belittles or calls into question the competencies of others a) isn't actually funny, and b) translates especially badly into electronic form, where it can easily be interpreted as harassment or bullying.
- Nearly everything that we do in Moodle, the Institute's Learning Management System, is automatically logged and easily trackable. Through Moodle, instructors can see exactly when

a student submitted an assignment, whether or not a student watched an instructional video, and whether or not a student looked at course materials posted online.

Institute Policies

Academic Integrity

No course assignment is worth staking your personal integrity over. Plagiarism, offering, and accepting solutions to assessments when a student is required to work individually or create their own solutions are cases of academic misconduct. These instances are taken seriously and will be reported. Anyone found cheating will not be permitted to withdraw and will be (appropriately) heavily penalized in the course.

It is expected that any work submitted for assessment represents the intellectual work of the individual(s) submitting the work. Any attempt to pass off the intellectual work of another (including the work generated by Large Language Models like ChatGPT) as their own or without proper attribution is an example of academic misconduct and is subject to the penalties described in the Rose-Hulman Academic Rules and Procedures and Student Handbook documents.

The Student Handbook and Rose-Hulman's Academic Rules and Procedures describe penalties and processes invoked as a consequence if academic misconduct (such as cheating, plagiarizing, or interfering with the academic progress of other students) takes place. It is the responsibility of each student to know and follow Rose-Hulman's rules about academic integrity.

Incompletes

Incompletes are granted only when a student is forced to miss several days of class due to extraordinary circumstances such as a documented confining illness or family emergency. To receive an incomplete grade, students should obtain approval from the professor before the last day of class.

No incompletes will be given unless a prior written agreement exists, following typical Rose-Hulman policy. In this course, an "I" grade will be given only in cases of extreme hardship. Poor performance in the course is not grounds for an incomplete. An incomplete shall be recorded as an "F" if the work is not completed within the time agreed upon by the professor and student.

Emergency Information

To receive email or text messages regarding emergency situations that may impact campus and, possibly, the delivery of classes, register for RAVE alerts and/or follow @Rose-HulmanAlert on Twitter. Any announcements about the Institute's ability to offer classes will be shared on Rose-Hulman's website.